

Quiz 3

● Graded

Student

Deva Pushkaran

Total Points

10 / 10 pts

Question 1

(no title)

4 / 4 pts

1.1 (no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

1.2 (no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0.5 pts Partial: Missing set notation

+ 0 pts Incorrect

1.3 (no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

1.4 (no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

Question 2

(no title)

1 / 1 pt

✓ + 1 pt Full, clear explanation

+ 0.5 pts Partial: Explanation has aspects that are correct but could be improved

+ 0 pts Missing or too incomplete to correctly address the issue

Question 3

(no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0.5 pts Partial correct explanation

+ 0 pts Incorrect

Question 4

(no title)

3 / 3 pts

4.1

(no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

4.2

(no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

4.3

(no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

Question 5

(no title)

1 / 1 pt

✓ + 1 pt Correct

+ 0 pts Incorrect

Question 6

(no title)

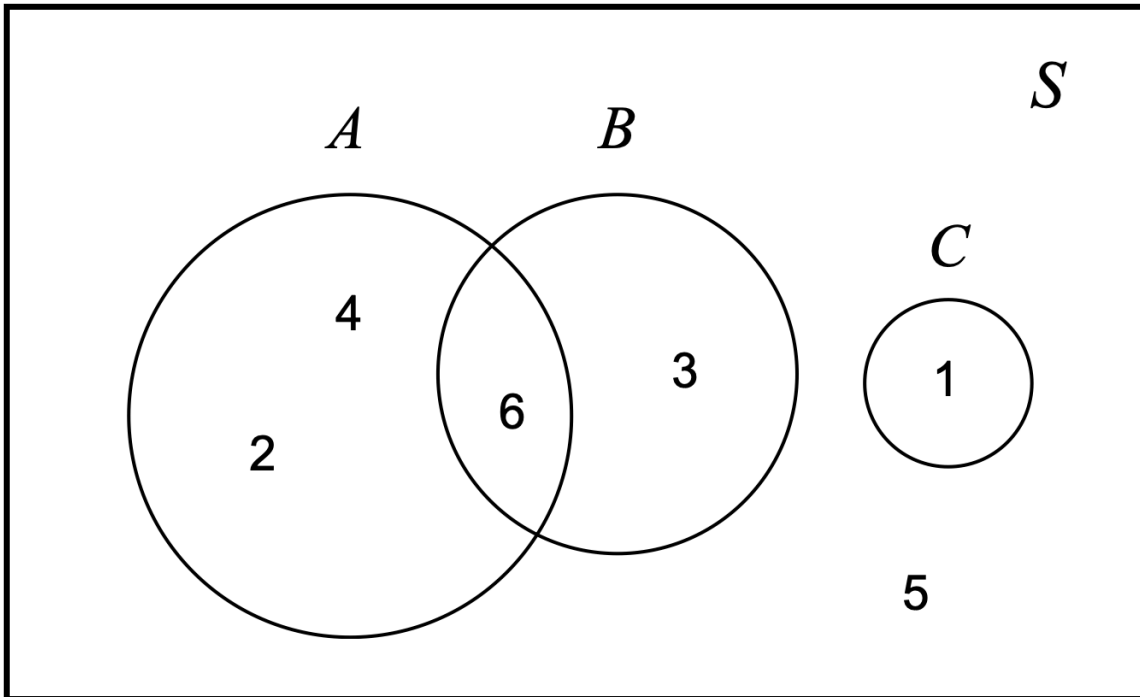
0 / 0 pts

✓ + 0 pts Incorrect

Q1

4 Points

Answer the following questions regarding the outcomes of rolling a fair die based on the Venn diagram below.



Q1.1

1 Point

Which of the following statements are true based on the given Venn diagram?

☒ A is the event that the outcome is an even number.

☐ A and B are disjoint.

☒ Every possible outcome is in the sample space.

☒ B and C are mutually exclusive.

Q1.2**1 Point**

What are the contents of $A \cap C$? Answer using set notation enclosed in $\{ \}$

$\{1,2,4,6\}$

Q1.3**1 Point**

What is $P(C)$, the probability of event C?

$1/6$

Q1.4**1 Point**

What is $P(B \cup C)$, the probability of events B or C?

$1/2$

Q2**1 Point**

Briefly describe what it means for two events to be mutually exclusive.

It means if there are 2 events then if one event occurs the other can not. ie event 1 happens event 2 can not. Event 2 happens event 1 can not.

Q3**1 Point**

How does the general Addition Rule in probability differ from our third axiom of probability?

The difference depends on the two events ability to be mutually exclusive.
The General Addition Rule is for calculating probability regardless if the two events may or may not be mutually exclusive.
The Third Axiom should only be used when trying to calculate the probability of two events that are mutually exclusive.

Q4**3 Points**

You have have two boxes, each with 2 red marbles and 1 green marble inside.

Q4.1**1 Point**

If you pull a single marble from one of the boxes, what is the probability of getting a red marble from that box? That is, what is $P(\text{red})$?

$2/3$

Q4.2**1 Point**

If you have two boxes and pull a marble from each, what is the probability of getting a red marble from the first box and a green marble from the second? That is, what is $P(\text{red from box 1}) \cap (\text{green from box 2})$?

$2/9 (2/3 * 1/3)$

Q4.3**1 Point**

What is the probability of drawing $P(\text{red from box 1}) \cup (\text{green from box 2})$?

$7/9$

Q5**1 Point**

In a simulated video game world, the days can be either sunny or cloudy, and the weather each day is independent of the weather any other day.

It only rains on cloudy days. There is a 60% chance a day is sunny. If a day is cloudy, there is a 30% chance there will be rain.

What is the probability that any given day will be rainy?

12% or $(4/10 * 3/10)$

Q6**0 Points**

Extra question (no points): Enter a number between 1 and 1000. Your goal is to select the lowest non-duplicated number in the class.

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